

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 90

SELF-OSCILLATING ECCENTRIC WEIGHT ALL-TERRAIN MOTORS

When the ground goes flat, carry your own hill
an offset weight rolls the ball from inside



a ground machine — the vet confused it, then corrected itself

CENTER-OF-GRAVITY DRIVE · LINKAGE-FREE
SHIFT THE WEIGHT — ROLL THE WORLD

Ground-contact propulsion — v1.0 in force

GP-IM-090

Revision v1.0 — 22 August 2026

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VETTED — 4-AI WEB — BATCH 16 — STANDS

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 90

PHASE 90: SELF-OSCILLATING ECCENTRIC WEIGHT CENTER-OF-GRAVITY ALL-TERRAIN MOTORS —

STATUS: CURRENT. When the ground goes flat, gravity quits — so the ball carries its own hill inside it. Phase 90 suspends a sealed spherical kinetic drive pod at the exact geometric center of the Phase 89 geodesic frame: a heavy offset counterweight on a low-power, high-torque motor (Phase 71 sodium-ion bank), continuously shifting the assembly's center of gravity so the ball rolls over dead-flat valleys, dunes, and uphill grades. No wheels, no steering rods, no exposed joints — the unibody rotation is the drivetrain. This is a GROUND machine, and the record is exact about it: internal eccentric masses cannot translate a free-floating vehicle — momentum cancels over the complete cycle — but against terrain, vibration and weight-shift do useful work. The vet confused this phase with the dumped Phase 37 mass-shifting energy system, was asked to its face, and corrected itself: Phase 90 STANDS.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-090, v1.0 — 22 August 2026. The ninety-first manual of the program — 91 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the eccentric-drive law as seated (contents line, the two design bodies — verbatim) — §2 why it works (ground contact versus free space — graded) — §3 the system in the pod — §4 the doctrine record — §5 the vet record (the confusion, traced and unwound) — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 90: **Self-Oscillating Eccentric Weight Center-of-Gravity All-Terrain Motors.** Propels heavy all-terrain cargo sleds across rugged planetary surfaces by sequentially shifting off-center, eccentric internal weights, creating cyclic center-of-gravity lurches to drive traction across loose gravel or slick ice.'

THE MASTHEAD LINE (verbatim): 'Workshop Toy Analog: Motorized Self-Rolling Children's Pet Balls & Internal Pendulum Gyros [extracted verbatim from the combined masthead line]'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Active Self-Propulsion Across Flat Extraterrestrial Terrain Baselines [*WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%'*] [extracted verbatim from the combined masthead line]'

ECCENTRIC CENTER-OF-GRAVITY DRIVES (VERBATIM)

* Core Assembly: Suspends a sealed, spherical kinetic drive pod in the exact geometric center of the Phase 89 expandable scissor-joint geodesic frame. * The Asymmetrical Pendulum Cam: Integrates a heavy, offset internal counterweight driven by a low-power, high-torque motor fueled by Phase 71 Sodium-Ion batteries. * Continuous Center Shift: Continuous rotation of the internal mass shifts the drone's center of gravity permanently outside its structural footprint, forcing the expanded sphere to perpetually fall forward into an active, high-velocity rolling velocity vector.

INDEPENDENT ALL-TERRAIN PROPULSION (VERBATIM)

* Flat Plain Traversal Mode: Overrides gravitational slope deceleration thresholds. The internal oscillating engine engages automatically upon transitioning onto dead-flat valleys, sand dunes, or uphill planetary grades. * Linkage-Free Kinematics: Eliminates fragile external wheels, steering rods, and exposed joints. The unibody rotation of the outer S-bend sugar-scoop shell provides [*WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%'*] enclosed mechanical propulsion, permanently insulating the drive from abrasive lunar dust sandblasting or gear-tooth shear lockups.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE FREE-SPACE BAR, AFFIRMED AND QUOTED: internal eccentric masses cannot translate a free-floating vehicle — momentum cancels over the complete cycle. The audit's own Register Entry 2 and the vet agree word for word. Phase 90 never claimed otherwise.

THE GROUND-CONTACT CASE, AFFIRMED — THE MACHINE'S TRUE PHYSICS: against terrain, a shifting center of gravity is propulsion — the ball falls toward its own offset weight, continuously, and the ground closes the momentum loop. The vet said it itself: 'for a ground-contact machine, vibration can absolutely interact with the terrain and produce useful motion. That's a legitimate mechanical class.' Self-rolling toys prove the class on every kitchen floor.

THE LINKAGE-FREE CHASSIS, AFFIRMED: no wheels, steering rods, or exposed joints means nothing to jam with dust, freeze, or shear — the failure modes of conventional rovers are deleted by omission, not managed by engineering.

THE DRIVE BUDGET, AFFIRMED AS SIZED: a low-power high-torque motor turning an offset mass is a small, steady draw — sized to the Phase 71 sodium-ion bank in the law text.

§3 — THE SYSTEM IN THE POD

- THE POD: sealed spherical kinetic drive suspended at the geometric center of the Phase 89 frame.
- THE WEIGHT: heavy offset counterweight — the asymmetrical pendulum cam — on a low-power, high-torque motor.
- THE BANK: Phase 71 sodium-ion cells feeding the drive; steady draw, no surge class.
- THE MODES: passive gravity tumble on slopes; the oscillating engine engages automatically on dead-flat valleys, dunes, and uphill grades.
- THE SHELL: unibody rotation — no wheels, rods, or exposed joints anywhere in the propulsion path.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE CARRY-YOUR-OWN-HILL DOCTRINE (seated with the pod): when the terrain stops helping, bring the slope with you — the offset weight is a portable grade.
- THE GROUND-VS-SPACE DOCTRINE (the batch-16 lesson, both audits agreed): the same internal mass that is fraudulent in free space is legitimate on the ground — the difference is the contact, and the record now says so in both audits' words.
- THE DELETE-THE-JOINTS DOCTRINE (seated with the chassis): what is not there cannot jam — the unibody is reliability by omission.
- THE TOY-TO-TOOL DOCTRINE (the masthead analog, verbatim): the motorized self-rolling pet ball and the internal pendulum gyro — the class was always real; the phase grows it up.

§5 — THE VET RECORD

THE PHASE 90 QUESTION, ASKED AND ANSWERED (batch 16, seated verbatim in the master): his question — 'why does it think its obsolete, is it confused?' The answer: YES — CONFUSED, AND TRACEABLY SO. The vet had read the author's Phase 37 ruling (verbatim in the record: 'this mass shifting shit not me, that drive was connected to the mayernick as additional energy, it was made obsolete, but not the main drive') — which killed the AI-origin flywheel/internal-mass-shifting ENERGY system attached to the Mayernick — and echoed it onto Phase 90, a different machine. The vet's free-space physics was right and irrelevant: Phase 90 is a ground machine. The vet opened vet batch 16 with its own correction: 'I've corrected Phase 90 in the running audit: current design, not obsolete. The

flywheel/mass-shifting issue belongs to the separate ship system you dumped.' Both audits now agree: Phase 90 STANDS as seated. Graded under the 4-AI vet web (sitting 565).

§6 — BUILD SEQUENCE

- STEP 1 — Build the pod: sealed spherical drive, offset mass on the high-torque motor; balance map logged across the full rotation.
- STEP 2 — Mate the frame: pod suspended at the geometric center of the Phase 89 geodesic shell; concentricity gauged.
- STEP 3 — Wire the bank: Phase 71 sodium-ion feed; draw profile benched across duty cycles.
- STEP 4 — Bench the traction: roll-out on flat, sand, and grade; traction versus surface class logged — the number this machine lives or dies by.
- STEP 5 — Prove the mode switch: automatic engagement on slope-to-flat transitions; hysteresis tuned so the drive never hunts.
- STEP 6 — Publish the ledger: traction curves, draw, duty cycle — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 89 — the roll-ball frame (GP-IM-089): the shell this motor drives; 89 owns the slopes, 90 owns the flats.
- PHASE 71 — the sodium-ion matrix (GP-IM-071): the drive bank.
- PHASE 37 — the Mayernick ruling (GP-IM-037): the killed mass-shifting ENERGY system this phase is not — the distinction both audits now enforce.
- PHASE 95 — the prism-lock feet (GP-IM-095): anchoring the harvester's work sites.

§8 — DO-NOT-BUILD

- NEVER sell it as free-space propulsion — internal masses cannot translate a free-floater; the ground is the machine's other half (Register Entry 2, both audits).
- NEVER resurrect the Phase 37 conflation — the dumped flywheel energy system and this ground drive are different machines; the record is exact.
- NO external drivetrain grafts — wheels, rods, and exposed joints are the failure modes this phase deleted.
- NEVER run the drive on the tumble slopes — gravity is free; the motor is for where gravity quits.

§9 — OWED

THE TRACTION BENCH, OWED: roll traction versus surface class (regolith, sand, sintered ice, plate rock); counterweight duty cycle and draw profile against the Phase 71 bank; mode-switch hysteresis figures.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-090, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). The ninety-first manual of the program — 91 of 290. Built 22 August 2026, third of the 20-manual 'clear the 100 mark' shift ('do another 20 while i away so we clear the 100 mark for the day when i get back'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564 and 566): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back'. The phase's own chain, all seated in the master: the contents line and the two design bodies (as seated); the 12 August naming batch (formerly 'SELF-OSCILLATING GEODESIC ALL-TERRAIN HARVESTER'); the Phase 37 conflation traced in batch 15 and unwound by the vet itself in batch 16 ('I've corrected Phase 90 in the running audit: current design, not obsolete', verbatim); the Q&A in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the traction bench, or his word.

END OF MANUAL — PHASE 90